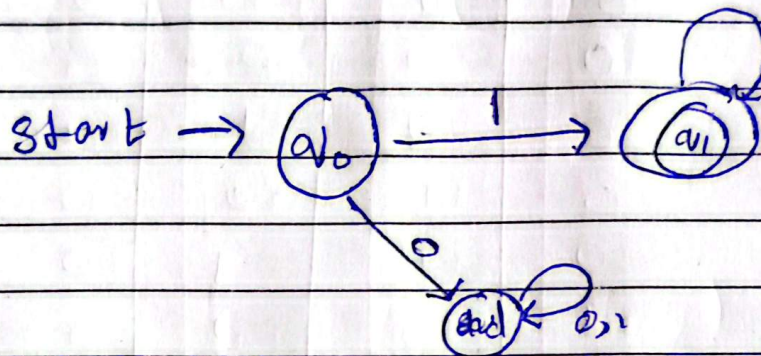


Q1) Task A)

regular expression = $1(0+1)^*$

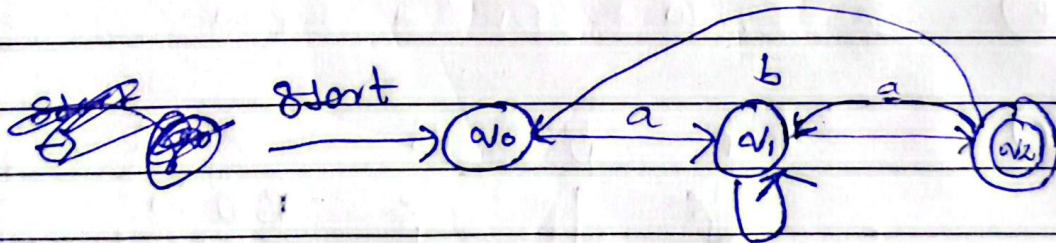
Task B)



Task C) accepted: 1011 ; rejected: 0111

Q2) Task A) regular expression $(a+b)^*ab$

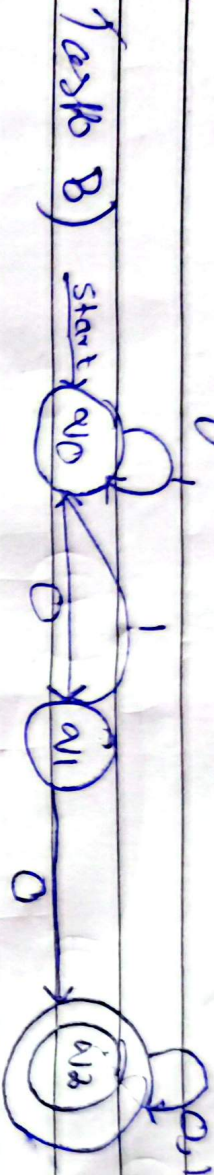
Task B) DFA:



Task C) ab, bbaab → accepted
rejected: ba abb

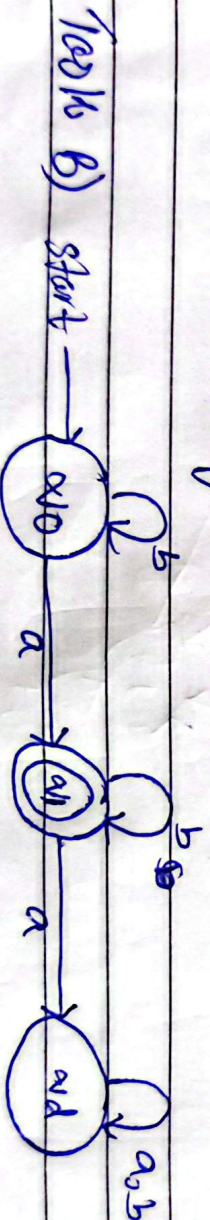
Date: _____

Q3) Task A) Regular expression: $(0+1)^*00(0+1)^*$



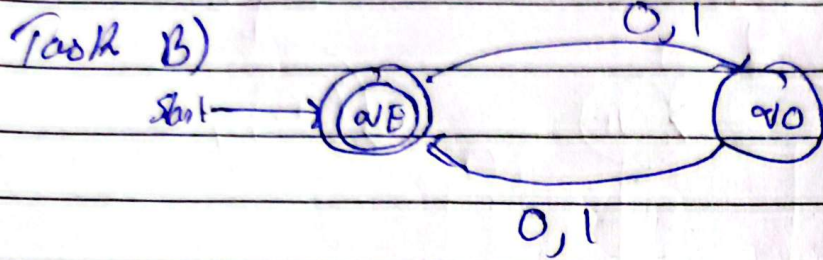
Task C) ~~accepted~~ accepted: 00 10011
rejected: 0101 1111

Q4) Task A) regular expression: b^*ab^*



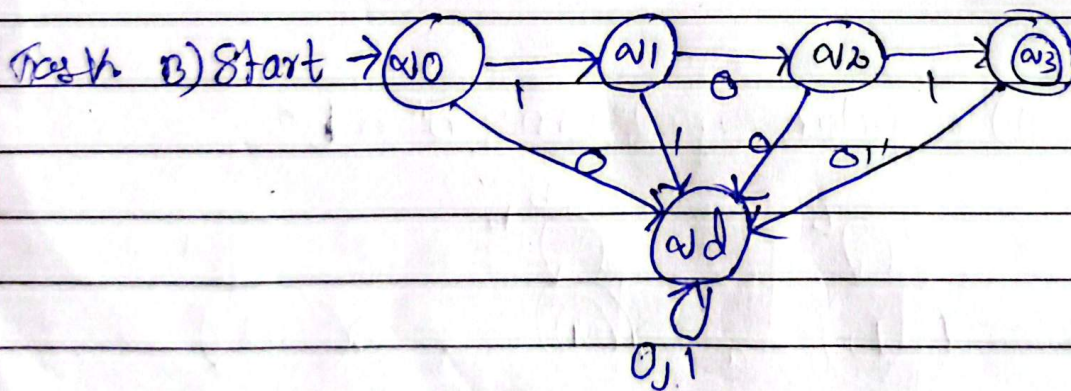
Task C) Accepted: a babb
rejected: bb aba

Q5) Task A) regular expression: $((0+1)(0+1))^*$



Task C) Accepted: ϵ (empty string), 1010
 rejected: 1, 010

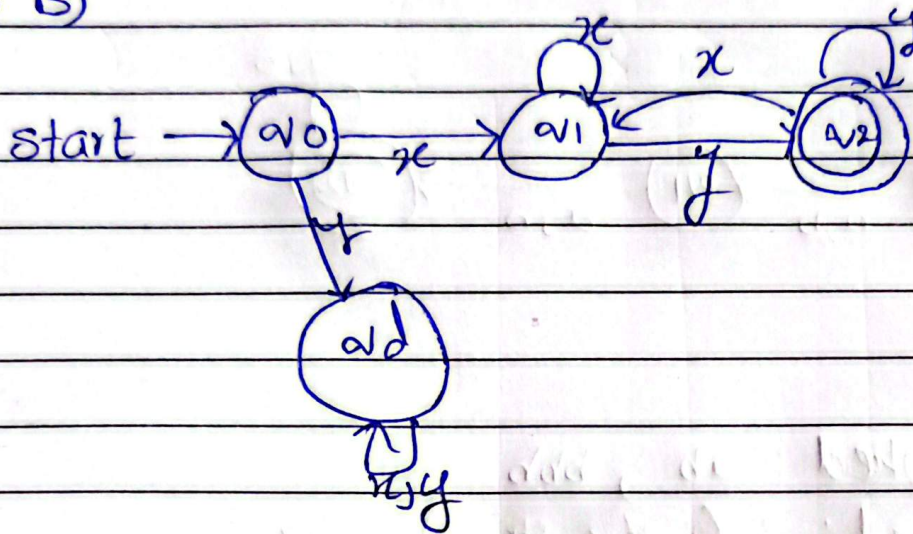
Q6) Task A) $\rightarrow$ regular expression: 101



Task C) accepted: 101
 rejected: 10 1011

Q 7) Task A) $x(x+y)^*y$ → regular expression

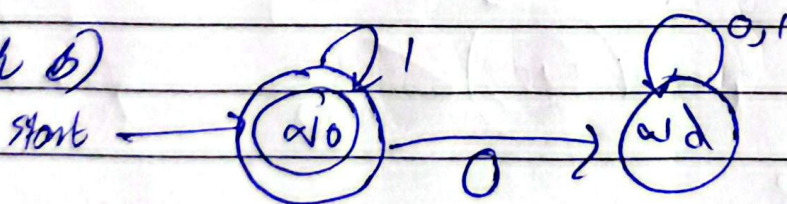
Task B)



Task C) Accepted: $xy, xnyxy$
 rejected: yx, xyn

Q 8) Task A) regular expression: 1^*

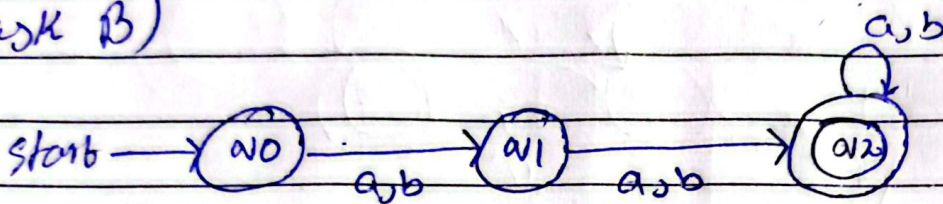
Task B)



Task C) Accepted: ϵ (empty string), 111
 rejected: $0, 10$

Q9) Task A) $(a+b)(a+b)(a+b)^*$

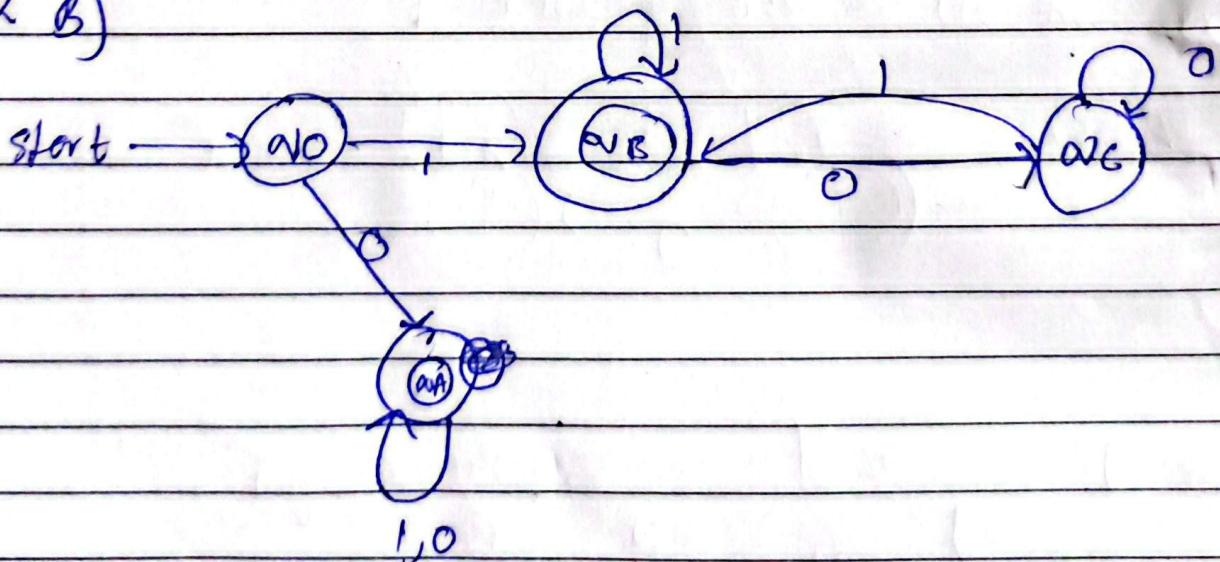
Task B)



Task C) Accepted: ab, bbb
 rejected: ϵ (empty string), a

Q10) Task A) regular expression: $0(0+1)^* + (0+1)^* \epsilon$

Task B)



Task C) Accepted: 0110 (Starts with 0), 101 (end with 1)
 rejected: 10, 1100